

STAINLESS STEEL MANUFACTURING · STEEL MELTING SHOP

AOD Operations

Process Analysis, Standardisation and Business Requirements

Portfolio edition. This is a sanitised version of work carried out in an operations role at a large stainless steel producer. The employer is not named, and all process parameters, quantities, temperatures, gas ranges, equipment identifiers and timings shown are illustrative placeholders, not operating values. The analysis structure, business rules, requirements, controls and validation logic reflect the work as performed. Nothing commercially sensitive is reproduced.

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About This Document

This document consolidates three pieces of process and requirements work carried out in the AOD (Argon Oxygen Decarburisation) area of a Steel Melting Shop. Each part addresses a distinct operational problem, and each follows the same structure: business problem, objectives and scope, stakeholders, current-state analysis, future-state design, rules and controls, and the measures used to confirm the change held.

The AOD converter is the point at which carbon is removed from the melt and the final chemistry of the heat is set. Material additions, gas ratios and temperature windows differ by grade family, and the process runs continuously across three shifts. The recurring problem across all three pieces of work is the same one: operating knowledge sat with experienced individuals rather than in a controlled reference, so execution varied by shift and by operator.

The work was performed in an operations role. The documentation approach, meaning structured requirements, business rules, traceability and acceptance criteria, is the approach used in business analysis, applied here to a manufacturing process rather than a software system.

Document Control

Field	Detail
Prepared by	S. Satyam Singh
Role at the time	Associate Manager, AOD Operations
Business area	Steel Melting Shop, AOD Operations
Coverage	Grade-wise addition pattern standardisation; AOD auto batching requirements; 5S material allocation
Methods applied	Process mapping (As-Is / To-Be), requirements definition, business rules, DMAIC, RACI, control planning, validation scenarios, traceability
Grade families	200 series, 300 series, 400 series
Version	2.0 (portfolio edition)

Contents

Part	Subject	Focus
A	Grade-Wise AOD Addition Pattern	Process mapping and standardisation of material addition sequence, temperature control and gas switch-over across three grade families
B	AOD Auto Batching	Business and functional requirements, bunker and weighing-hopper logic, exception handling and validation scenarios for automated material batching
C	5S Material Allocation at AOD	Current-state analysis, gap analysis and requirements for material location control, receipt handover and space utilisation

Terms Used

Term	Meaning
AOD	Argon Oxygen Decarburisation, the converter process used to refine stainless steel
Main blow / DB1 / DB2	Primary oxygen blowing stage, followed by second and third blow stages
Zero batch	Initial flux charge added before pouring
Dololime / Lime / Flux	Slag-forming additions used to control chemistry and protect refractory
Ferroalloys	Chromium, nickel and manganese bearing additions used to achieve target chemistry
MHS	Material Handling System: conveyors, bunkers, feeders and chutes serving the converter
CH1 / CH2	Charging hoppers feeding material into the converter
Weighing hopper	Intermediate hopper that weighs material before transfer to the charging chute
Reduction	Post-blow stage where chromium is recovered from slag

PART C

5S Material Allocation at AOD

Current-state analysis, gap analysis and business requirements for material location control, receipt handover and space utilisation in the AOD material allocation area.

C1. Background

This work addressed 5S implementation for material allocation in the AOD area of the Steel Melting Shop. The starting requirement was to improve visual management and reduce material wastage. Delivery covered process review, stakeholder discussion, material location tagging, bin allocation, receipt handover control and improvement of material handling practice.

The analysis required understanding the current operating process, identifying pain points, capturing stakeholder requirements, defining business rules and documenting the future process for shop-floor execution.

C2. Business Problem

Problem area	Business problem
Material receipt delay	Receipt information could carry over between shifts, creating confusion at handover
Visual management	Material locations were not readily identifiable from the control room or shop floor
Space utilisation	Long-stored material and unplanned placement reduced available working space
Small quantity material handling	Low-volume additives required better sub-bin allocation
Communication gap	Melters, operators, technicians and helpers needed a clear location reference for each material
Grade-wise equipment readiness	Chute requirement varied by grade family, so availability needed to match the running heat plan
Ownership gap	Material unloading and placement required clear responsibility, particularly on night shift

C3. Objectives

Objective	Description
Improve visual management	Create clear identification and tagging for material locations
Reduce material wastage	Improve material allocation discipline and reduce incorrect placement
Improve shift handover	Ensure receipt and material information is available before the next shift takes over
Improve material handling	Define bins, sub-bins and standard locations for frequently used materials
Improve communication	Allow the control room to communicate material location clearly to technicians and helpers
Improve process control	Align material booking and chute availability with the heat plan and grade requirement
Sustain the improvement	Create controls so the improved process can be maintained over time

C4. Scope

In scope	Out of scope
Material allocation in the AOD area	Supplier commercial negotiations
Tagging and nomenclature for material locations	Change to metallurgical grade design
Bin and sub-bin division for small quantity materials	Major civil construction work
Receipt handover and material identification control	ERP or software implementation
Material booking alignment with the heat plan	Automation of the weighing bridge system
Chute availability by grade requirement	Plant-wide 5S implementation outside AOD

C5. Stakeholders

Stakeholder	Need or interest	Engagement approach
Production Manager	Improved productivity and reduced wastage	Establish business priority and success measures
Shift-In-Charge	Clear material availability and handover information	Capture shift-level pain points and execution needs
Melter / Operator	Quick identification of required material	Understand real user workflow from the control room
Technician / Helper	Clear physical location of materials	Capture location communication issues
Store / Bunker Yard Team	Correct unloading and allocation reference	Define receipt and placement coordination
Material Unloading Supervisor	Ownership of unloading and correct placement	Define responsibility and handover control
Maintenance / Mechanical Team	Space allocation and equipment placement	Capture constraints on shop-floor movement and space

C6. Method

The work followed a DMAIC structure, with the analysis and documentation produced at each phase set out below.

Phase	Activity	Output
Define	Established the need to perform 5S of material allocation at AOD to improve visual management and reduce wastage	Problem definition, scope and objectives
Measure	Identified the need for nomenclature, bin division and a material allocation mechanism	Current state review, data points, process observation
Analyse	Reviewed current trends and shop-floor issues affecting material allocation	Gap analysis, root cause review, stakeholder discussion
Improve	Proposed material handling improvement, tagging, sub-bins and receipt control	To-Be process design, business requirements, options
Control	Defined how 5S at AOD would be implemented and maintained	Control plan, ownership, KPIs, process governance

C7. Pain Points and Gap Analysis

Pain point	Operational impact	Required change
Receipt handover delay	Shift-to-shift information delay created confusion over material receipt	Increase receipt copies and include shift, timing and chemistry detail
Unclear location identification	Operators and helpers could not identify material locations quickly	Tag all material locations with permanent, visible labels
No sub-bin system for small materials	Low-volume additives were harder to locate and control	Divide bins into sub-bins for low-volume additives
Material held for long periods	Space remained blocked by slow-moving material	Clear long-stored material gradually and set maximum quantity limits
Imprecise material booking	Excess or incorrect material availability affected space and planning	Book material from stores and bunker yard according to the heat plan
Chute availability not grade-aligned	Unnecessary chutes occupied space while required chutes could be unavailable	Vary chute availability based on the running grade family
Incorrect material placement	Material placed at the wrong location caused execution issues	Assign a material unloading supervisor with material knowledge

C8. Business Requirements

ID	Business requirement	Priority
BR-C01	The AOD material allocation area must have clearly tagged locations for each identified material category	High
BR-C02	Material bins must be divided into sub-bins for low-volume additives	High
BR-C03	Filled material trucks must carry an additional receipt copy to support shift handover	High
BR-C04	Receipt detail must include shift, timing and chemistry detail where applicable	High
BR-C05	Material must be booked from stores and bunker yard precisely according to the heat plan	High
BR-C06	Chute availability must be adjusted to the running grade family	Medium
BR-C07	Slow-moving materials must be cleared gradually to release space	Medium
BR-C08	Quantity of bulk material held in the area must be controlled to avoid unnecessary space blockage	Medium
BR-C09	Standard weights used for weighing bridge calibration must be tagged	Medium
BR-C10	A material unloading supervisor must be available in the receiving bay to ensure correct placement	High

C9. Functional Requirements

ID	Functional requirement
FR-C01	The material allocation process must support location identification from the control room using tagged location references
FR-C02	The process must allow operators to communicate material location to technicians and helpers without ambiguity
FR-C03	The bin structure must support separate storage of small quantity materials
FR-C04	The receipt handover process must ensure material receipt status is available before shift handover
FR-C05	The material unloading process must include verification by a responsible person before final placement
FR-C06	The process must support grade-wise chute planning based on the running heat plan

C10. Business Rules

Rule ID	Business rule
BU-C01	Each material category must be stored only in its identified tagged location
BU-C02	Small quantity materials must be placed in defined sub-bins and not mixed with bulk materials
BU-C03	Material receipt handover must include receipt count, shift, unloading time and chemistry information where applicable
BU-C04	Material must be booked according to the heat plan rather than to excess availability
BU-C05	Chute availability must match the grade plan for the running series
BU-C06	Chute requirement must be reduced where the running series permits, to avoid unnecessary space usage
BU-C07	Long-stored materials must be reviewed and cleared gradually to create space
BU-C08	The material unloading supervisor must ensure material is not placed at the wrong location
BU-C09	Tagging must be visible enough for store, bunker yard and AOD team members to identify locations

C11. Acceptance Criteria

ID	Acceptance criterion
AC-01	Given a material truck is unloaded, when shift handover happens, then the next shift receives clear receipt information without confusion
AC-02	Given a melter or operator requests a material location, when the control room checks the tagged layout, then the location can be communicated clearly to the technician or helper
AC-03	Given small quantity materials are received, when they are stored, then they are placed in the correct sub-bin
AC-04	Given a heat plan is available, when material is booked from stores or bunker yard, then booking matches the heat plan requirement
AC-05	Given a grade family is running, when chutes are arranged, then only the required number of chutes occupies the area
AC-06	Given a 5S audit is performed, when the material area is reviewed, then material tags, bin allocation and space clearance are visible and traceable

C12. RACI Matrix

Activity	Analyst	Shift-In-Charge	Store / Bunker Yard	Unloading Supervisor	Production Manager
Capture current process and pain points	R	C	C	C	A
Define material tagging requirement	R	C	C	C	A
Prepare bin and sub-bin allocation logic	R	C	C	C	A
Implement receipt handover control	C	A	R	R	C
Ensure correct material placement	C	A	C	R	C
Monitor 5S sustainment	C	R	C	R	A

R = Responsible · A = Accountable · C = Consulted · I = Informed

C13. Measures

Measure	Measurement method	Target
Material location identification time	Time taken to identify material location from control room or shop floor	Reduced search time
Incorrect placement incidents	Count of materials found in an incorrect location	Reduced incidents
Receipt handover accuracy	Receipt count and information available during shift handover	Improved handover accuracy
Tagged location completion	Number of material locations with visible tags	All critical locations tagged
Sub-bin readiness	Availability of sub-bins for small quantity materials	Defined sub-bins available
Space availability	Review of space blocked by slow-moving material	Improved space usage
5S compliance score	Periodic 5S audit of the AOD material allocation area	Sustained compliance

C14. Benefits

Area	Benefit
Visual management	Material locations become easier to identify for the AOD team, store and bunker yard
Communication	Melter or operator can direct technician and helper using clear location references
Shift handover	Receipt information becomes clearer between shifts
Space management	Long-stored and excess material can be controlled to improve working space
Material control	Small quantity materials are separated and easier to track
Operational efficiency	Less time lost searching for, clarifying and correcting material placement
Sustainment	Control measures support maintenance of 5S after implementation

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